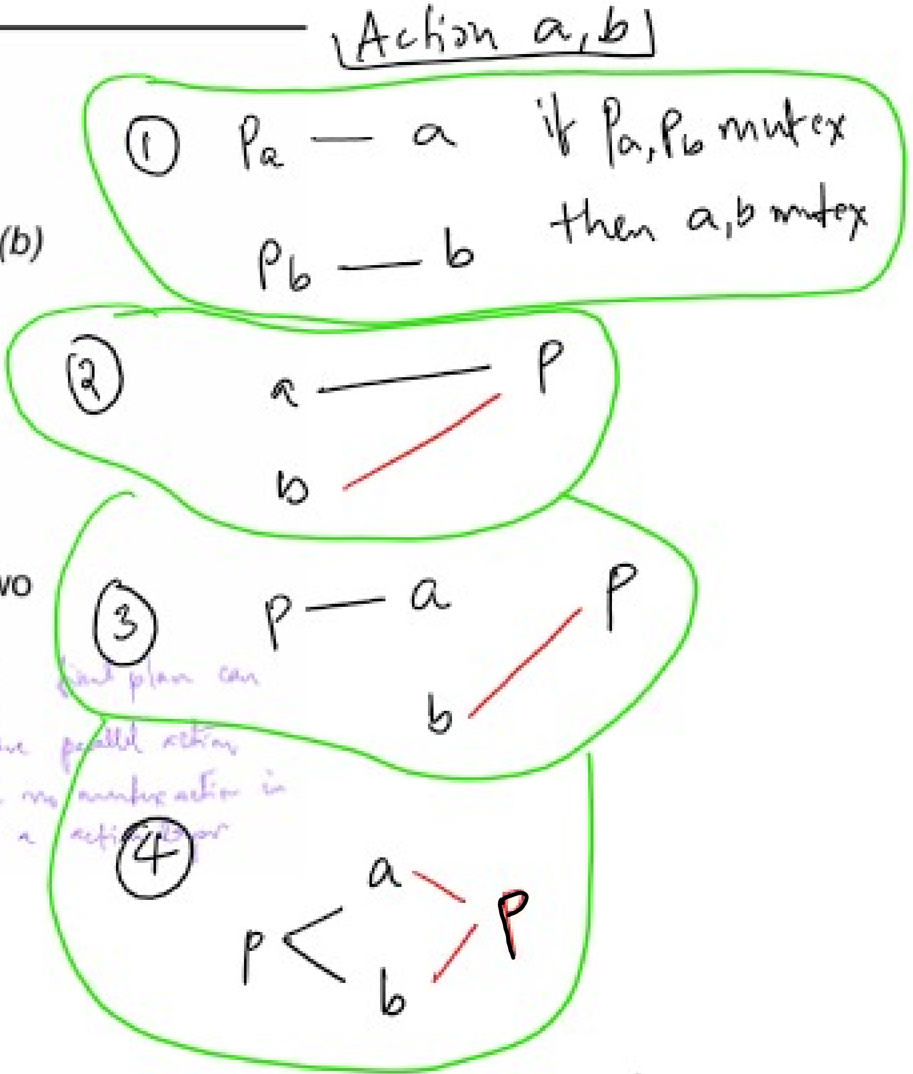
Mutual Exclusion (mutex)

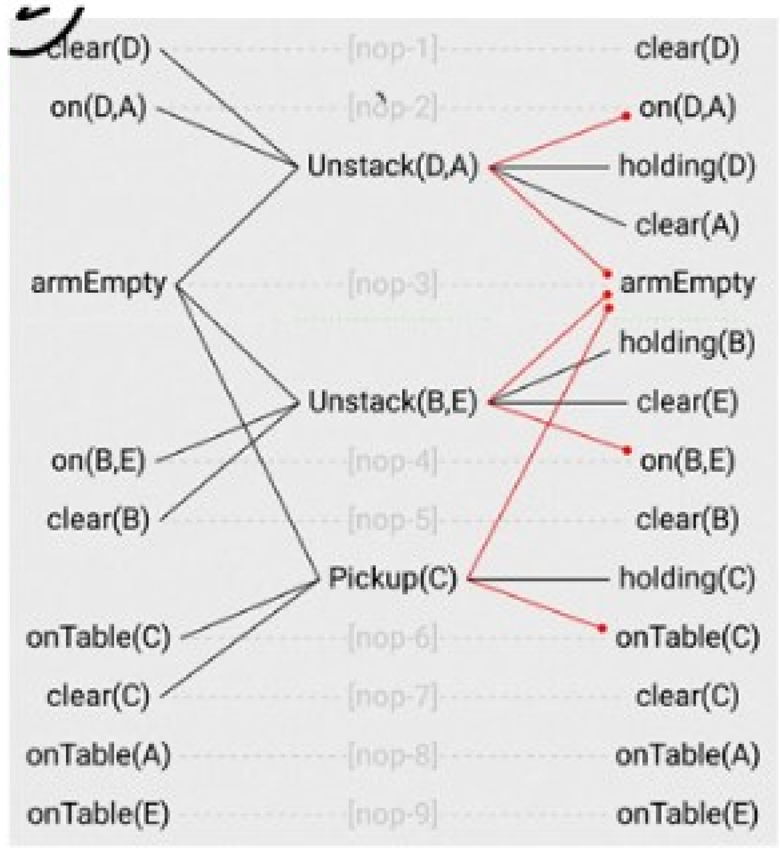
both can't happen together

Two actions $a \in A_i$ and $b \in A_i$ are *mutex* if one of the following holds,

1. *Competing needs*: There exist propositions $p_a \in pre(a)$ and $p_b \in pre(b)$ such that p_a and p_b are mutex (in the preceding layer).
2. *Inconsistent effects:* There exists a proposition p such that $p \in effects^+(a)$ and $p \in effects^-(b)$ or *vice versa*. The semantics of these actions in parallel are not defined. And if they are linearized then the outcome will depend upon the order.
3. *Interference:* There exists a proposition p such that $p \in pre(a)$ and $p \in effects^-(b)$ or *vice versa*. Then only one linear ordering of the two actions would be feasible. a → b so can't parallel.
4. There exists a proposition p such that $p \in pre(a)$ and $p \in effects^-(a)$ *and also* $p \in pre(b)$ and $p \in effects^-(b)$. That is the proposition is consumed by each action, and hence only one of them can be executed. always there exists

final plan can have parallel action but no. of action in a action layer





Two propositions $p \in P_i$ and $q \in P_i$ are mutex if *all* combinations of actions <a,b> in A_i with $p \in effects^+(a)$ and $q \in effects^+(b)$ are mutex.

☆ Two propositions are mutex only if all pairs of all actions adding them to the layer are mutex.

☆ An action is applicable in Graphplan when –

- All preconditions are in the previous layer
- No pair of preconditions is mutex
  (All pairs of pre-cons are non-mutex)